

CloudX Demo Scripts

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Prep/Prerequisite

Must have Ollama installed on laptop and configured.

Must have Powershell and use in Admin role.

Ensure all environment is set in PATH for Postgres, Ollama, Flyway

I've configured mine to use the postgres user, but you could change this for enhanced security.

Ensure Postgres is running and has a demo database. If needed, set `$env:PGPASSWORD`.

Unzip project; ensure psql and **flyway are on PATH** (or set `FYW_EXE` in scripts).

```
createdb -h localhost -U postgres demo
```

```
$env:PGPASSWORD = "use password"
```

The `pg_stat_statements` extension must be installed/configured

After downloading the scripts, you may have to unblock the scripts in each of the file properties due to security in Windows.

Pipeline (Governance Migrate Smoke Observe)

```
.\scripts\run.ps1
```

Governance: verify-hash.ps1 checks SHA-256 of migrations.

Migrate: Flyway applies sql/V1..V4 (GitOps-style versioned migrations).

Smoke: Confirms objects exist and RLS is enabled on invoices.

Observe: Stage timings; tries `pg_stat_statements` (skips if not permitted).

2) Seed & Prove RLS

```
.\scripts\seed.ps1
```

This is the full code of what will be done in the Canary test if we were to do it in reality... 😊

```
psql -h ${HostName} -p $Port -U $User -d $Database -c "ALTER TABLE
public.customers ENABLE ROW LEVEL SECURITY; ALTER TABLE
public.customers FORCE ROW LEVEL SECURITY;"
```

```
psql -h ${HostName} -p $Port -U $User -d $Database -c "ALTER TABLE
public.invoices ENABLE ROW LEVEL SECURITY; ALTER TABLE public.invoices
FORCE ROW LEVEL SECURITY;"
```

```
psql -h ${HostName} -p $Port -U $User -d $Database -c "CREATE POLICY
rls_customers_tenant ON public.customers USING (tenant_id =
current_setting('app.tenant_id', true));"
```

```
psql -h ${HostName} -p $Port -U $User -d $Database -c "CREATE POLICY
rls_invoices_tenant ON public.invoices USING (tenant_id =
current_setting('app.tenant_id', true));"
```

- Policy uses current_setting('app.tenant_id'); helper set_tenant() sets per session.
- Least-privilege role app_readonly granted SELECT only.

Progressive Delivery via Feature Flag (Canary)

```
.\scripts\toggle-canary.ps1 -Enable:$true
```

Script will enable Canary

Set the RLS

Check the views

Report back with a 1 if valid

```
.\scripts\toggle-canary.ps1 -Enable:$false
```

Will disable Canary once complete

- V3 adds v_invoices_current with 5% discount when demo.canary_enabled=on.
- V4 reverts via roll-forward (safe rollback pattern).

Optional CI & GitOps (2 min)

CI: review `.github/workflows/ci.yml` (Postgres service integrity migrate smoke canary).

```
act -W .github/workflows/ci.yml -j ci --secret PGPASSWORD=postgres
```

GitOps (preview only):

```
.\ops\scripts\render-kustomize.ps1 -Overlay stable
```

```
.\ops\scripts\render-kustomize.ps1 -Overlay canary
```

Troubleshooting

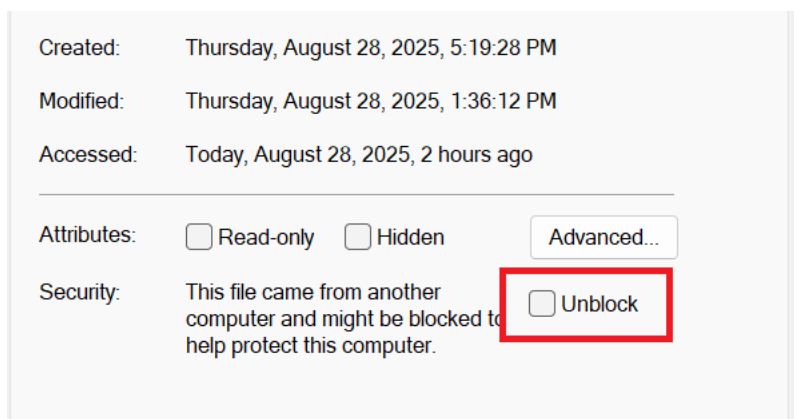
Auth: set `$env:PGPASSWORD` or pass `-Password` to scripts.

Flyway not found: set `FYW_EXE` in `scripts\run.ps1`.

pg_stat_statements warning: needs superuser; safe to ignore in demo.

Custom port: pass `-Port 5432` to scripts.

“ps1 is not digitally signed”: Right click on each file to run these powershell scripts in your own secluded test environment and choose to “unblock” from the general properties:



File Map

- `sql/V1..V4` schema, RLS, feature-flag view, revert
- `scripts/verify-hash.ps1` integrity guardrails
- `scripts/ai-review.ps1` optional Ollama LLM review
- `.github/workflows/ci.yml` CI mirrors local pipeline
- `app/k8s/overlays/*` stable vs canary (Kustomize)